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3. n = 4

l = 0, 1, 2, 3

$\therefore$ 4 sub shells s, p, d & f.

$l = 0$ $m_l = 0 \Rightarrow$ one 4s orbital.

$l = 1$ $m_l = -1, 0, +1 \Rightarrow$ three 4p orbitals.

$l = 2$ $m_l = -2, -1, 0, +1, +2 \Rightarrow$ five 4d orbitals.

$l = 3$ $m_l = -3, -2, -1, 0, +1, +2, +3 \Rightarrow$ seven 4f orbitals.

Over all 16 orbitals are possible.

4.

Orbital	**n**	***l***	**Radial node n – *l* – 1**	**Angular node *l***	**Total node n – 1**
3d	3	2	0	2	2
4f	4	3	0	3	3

5. $E_n = \frac{-13.6}{n^2} eV$

Second excited state

$n = 3 \quad \therefore E_3 = \frac{-13.6}{9} eV$

$E_3 = -1.51 eV$

6. Electronic configuration of Fe^{3+} is $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^0\ 3d^5$

Five unpaired electrons.

1	1	1	1	1

$3d^5$

Electronic configuration of Mn^{2+} is $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6\ 4s^0\ 3d^5$

$\therefore$ **Five unpaired electrons.**

Electronic configuration of Ar: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^6$

no unpaired electrons.

7.

